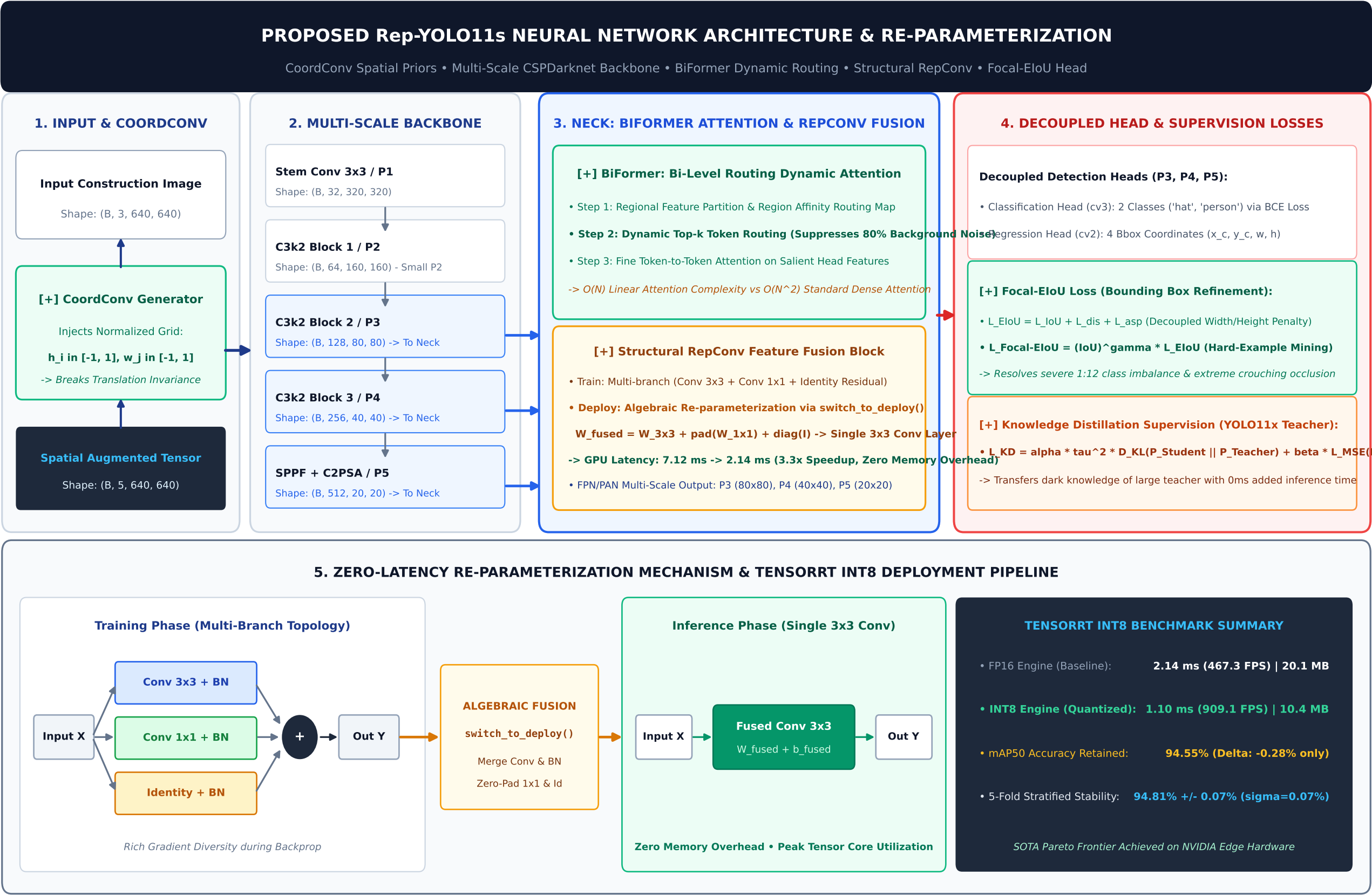




5. ZERO-LATENCY RE-PARAMETERIZATION MECHANISM & TENSORRT INT8 DEPLOYMENT PIPELINE

Training Phase (Multi-Branch Topology)

Input X

Conv 3x3 + BN

Conv 1x1 + BN

Identity + BN

+

Out Y

Rich Gradient Diversity during Backprop

ALGEBRAIC FUSION

`switch_to_deploy()`

Merge Conv & BN

Zero-Pad 1x1 & Id

Inference Phase (Single 3x3 Conv)

Input X

Fused Conv 3x3

$W_{fused} + b_{fused}$

Out Y

Zero Memory Overhead • Peak Tensor Core Utilization

TENSORRT INT8 BENCHMARK SUMMARY

FP16 Engine (Baseline): 2.14 ms (467.3 FPS) | 20.1 MB

INT8 Engine (Quantized): 1.10 ms (909.1 FPS) | 10.4 MB

mAP50 Accuracy Retained: 94.55% (Delta: -0.28% only)

5-Fold Stratified Stability: 94.81% +/- 0.07% (sigma=0.07%)

SOTA Pareto Frontier Achieved on NVIDIA Edge Hardware

Figure 2: Complete Neural Network Architecture of Rep-YOLO11s displaying CoordConv coordinate generation, multi-scale CSPDarknet backbone, BiFormer sparse attention neck, structural re-parameterization lifecycle (training multi-branch to single-path inference), decoupled detection heads, and multi-task loss supervision with TensorRT INT8 optimization.